DO NOW — RECALL

# Lesson 047 · Two- and Three-dimensional Arrays in C#

Name: _____ Class: _____ Date: _____

**1. How does index 0 work in C# arrays?**

**2. What is a property on a C# object?**

**3. What is the distinction between an array reference and an object instance?**

**4. A zero-indexed array a has 4 rows and 3 columns. Every element starts at 0. A loop runs r=0 to 3 inclusive and sets a[r,2]=r+1. Determine the final values in the last column, from top to bottom.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It accesses the first position.

### 2. Model answer

A member exposing data through get/set accessors.

### 3. Model answer

A reference points to the object; the object holds state.

### 4. Model answer

1, 2, 3, 4. Column index 2 is the last column; each row receives r+1.

### By the end, you can…

• A rectangular C# 2D array uses [row,column]. Example seats has 2 rows and 3 columns.

• A 3D array uses [layer,row,column]. new int[2,3,4] holds 24 integers; its last valid index is [1,2,3].

• Each dimension starts at zero; distinguish [1,2,3] from a single flat offset and never exceed a dimension's length.

### Vocabulary — know these terms

dimension — An indexing axis of an array.

row — First index in 2D example.

column — Second index in 2D example.

voxel — 3D grid element.

NEW LEARNING

# Read, annotate and explain

## Observe

A rectangular C# 2D array uses [row,column]. Example seats has 2 rows and 3 columns.

## Explain

A 3D array uses [layer,row,column]. new int[2,3,4] holds 24 integers; its last valid index is [1,2,3].

## Apply

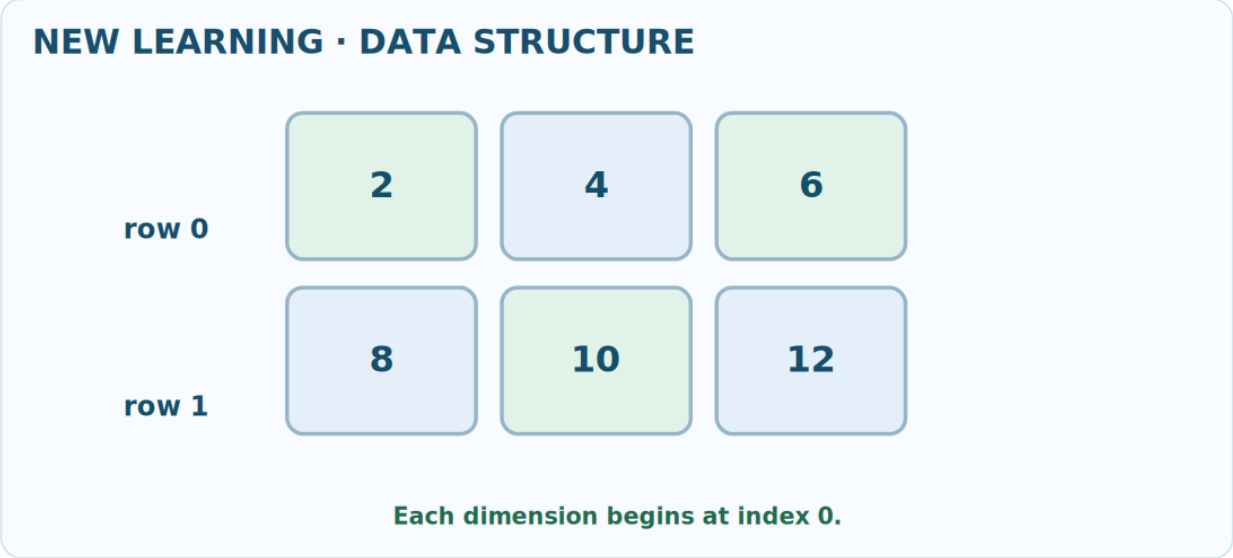
Each dimension starts at zero; distinguish [1,2,3] from a single flat offset and never exceed a dimension's length.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




NEW LEARNING · DATA STRUCTURE: 2, 4, 6, 8, 10, 12

I DO — WORKED EXAMPLE

# Follow the worked method

### Find seats[0,2] in {{3,5,7},{9,11,13}}.

### Trace and justify

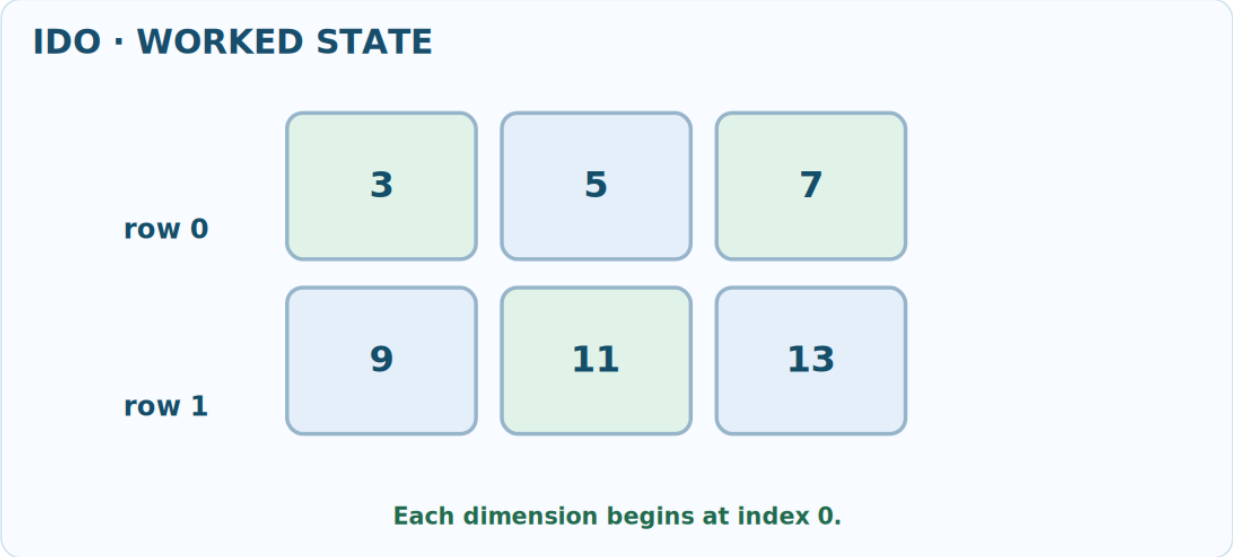
First row, third column gives 7. The column index 2 is valid because there are three columns.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Find seats[0,2] in {{3,5,7},{9,11,13}}.**

First row, third column gives 7. The column index 2 is valid because there are three columns.




IDO · WORKED STATE: 3, 5, 7, 9, 11, 13

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Work out final value from a different 3D array.
Source code / input:

```
int[,,] cubes=new int[2,2,3];
cubes[1,0,2]=14;
Console.WriteLine(cubes[1,0,2]);
```

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Find a[1,1] in {{2,4,6},{8,10,12}} and show indexes. [4]

### Question 2

A 3D array has sizes [2,3,4]. Calculate items and last index triple. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Write two nested C# loops that print all values in int[,] a of 3 rows and 2 columns, explaining order. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**In int[,,] x=new int[2,3,4], x[2,0,0] is valid.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

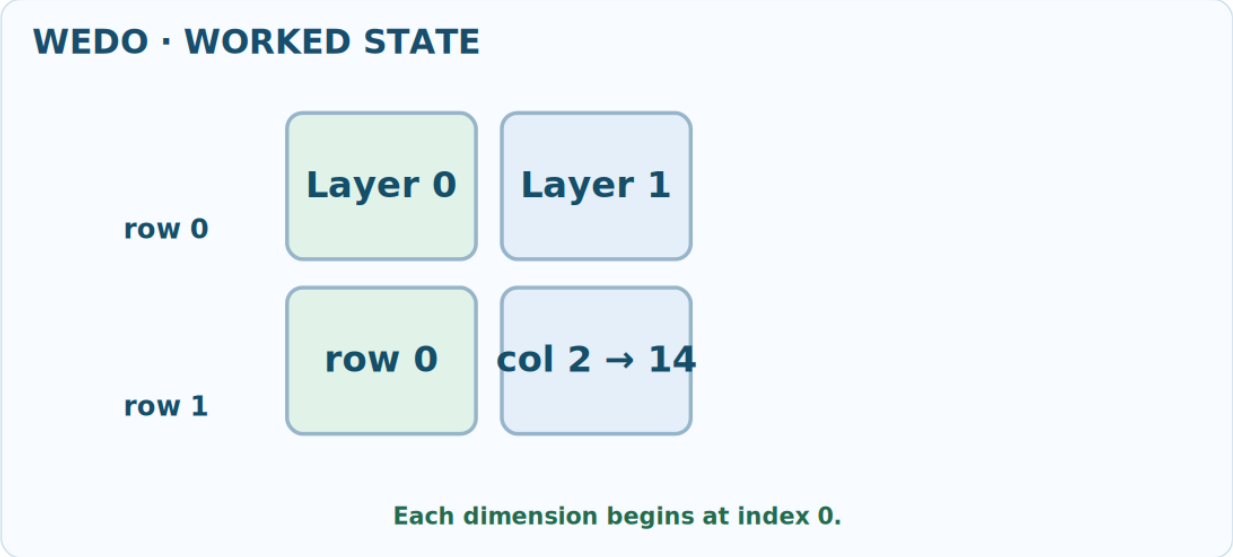
Trace and justify
Output 14: second layer, first row, third column. Six values per layer, twelve total.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Work out final value from a different 3D array.**

Output 14: second layer, first row, third column. Six values per layer, twelve total.




WEDO · WORKED STATE: Layer 0, Layer 1, row 0, col 2 → 14